

K-means Clustering of  
UNSW-NB15 Network  
Traffic: A From-Scratch  
Implementation

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### Abstract:

The implementation of K-Means Algorithm from scratch and clustering a classbalanced sample of UNSW-NB15.A

diagonal-Mahalanobis extension was implemented but run with identity weights

### Problem & Data Set:

We cluster network flows from the NSW-NB15 partitioned training set. It was obtained from the sharepoint link since the google drive link did not pan out. See the data/readme.md for more details. A class-balanced cap of 4,000 yields 3,730 rows. We use 39 standardized numeric features. Labels were used only for the sampling design and evaluation, never as clustering features. Some column will nulls and mismatched datasets were excluded

### Implementation:

K-means follows Lloyd's algorithm with random initialization. A restart stops when the maximum centroid shift is below  $10^{-4}$ , the labels stop changing, or 300 iterations are reached. An empty cluster is re-seeded at the point with the largest squared distance from that cluster's previous centroid. Against a scikit-learn reference at  $k = 10$ , the inertia ratio is 0.9966 and ARI is 0.9005. At other  $k$  values, scratch inertia was sometimes higher than the reference, likely from local optima or non-greedy seeding; this was not diagnosed further

### Choosing $k$ :

Over  $k = 2-15$ , silhouette in the Euclidean space peaks at  $k = 10$ .

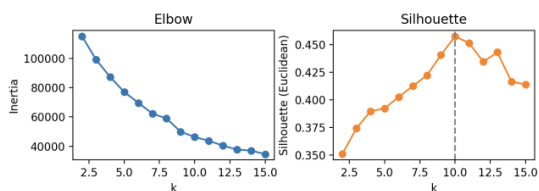


Table 1: Results at  $k = 10$  (identical for both runs; identity weights).

| Metric                     | Euclidean | Mahalanobis |
|----------------------------|-----------|-------------|
| Silhouette (Eucl. geom.)   | 0.4577    | 0.4577      |
| Silhouette (config. geom.) | 0.4577    | 0.4577      |
| Calinski-Harabasz          | 885.13    | 885.13      |
| Davies-Bouldin             | 0.8760    | 0.8760      |
| Homogeneity                | 0.2238    | 0.2238      |
| Completeness               | 0.3766    | 0.3766      |
| V-measure                  | 0.2807    | 0.2807      |
| Adjusted Rand              | 0.1152    | 0.1152      |
| Accuracy                   | 0.3271    | 0.3271      |
| Macro precision            | 0.3560    | 0.3560      |
| Macro recall               | 0.3107    | 0.3107      |
| Macro F1                   | 0.2522    | 0.2522      |

As we can see, Inertia decreases with a smooth elbow. The sweep used a single seed, so stability was not checked. The fact that there are **10 classes** is a **coincidence**. There is no relation with  $k$  and the labels.

### Evaluation:

Table 1 reports results at  $k = 10$ . Majority-vote accuracy, precision, recall, and F1 are in-sample cluster agreement, not held-out detection performance. Silhouette is computed in the common standardized Euclidean space.

### Euclidian vs Malhotrans:

The diagonal  $C$  was the identity (all 39 weights 1.0, in the feature order of feature names in results/metrics.json), so the Mahalanobis run reproduces the Euclidean run exactly. **No primary criterion was declared and no weighting experiment was run**, so no claim is made about whether weighting helps or hurts. Sample,  $k$ , seed, and restart budget were held fixed. Inertia is not comparable across distance metrics.

### Conclusion:

The from-scratch implementation matches the reference closely at  $k = 10$ , and the chosen  $k$  is supported by an interior silhouette peak. AI assistance used in this project is acknowledged in submission.md.

